Lab 7 – Fuzzing for Input Validation Bugs

Module: customs_duty.py (parse_declaration, duty_amount, hs_chapter) · Tools: Python 3.13, Hypothesis 6.170.15, pytest 8.3.3 · Repo: Mukilskanda/L6_L7_35_customs_duty (lab7)

Step 1 – Install dependencies

`pip install hypothesis pytest` → Successfully installed hypothesis-6.170.15

Step 2 – Run the original (buggy) script: `python customs_duty.py`

```
PS C:\Users\mukil\L6_L7_35_customs_duty\lab7> python $env:TEMP\orig_customs_duty.py
==== Running parse_declaration ====
Enter category:value (e.g. electronics:25000): electronics:25000
Category and Value: ('electronics', 25000.0)

==== Running duty_amount ====
Enter value@rate (e.g. 25000@20): 25000@20
Duty Amount: 5000.0

==== Running hs_chapter ====
Enter an HS code (e.g. 8517.12): 8517.12
Chapter: 85
```

Step 4 – Fuzz tests against the original code: 11 failed, 3 passed

```
test_customs_duty.py:137: AssertionError
=========================================================== short test summary info ============================================================
FAILED test_customs_duty.py::test_parse_declaration_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_customs_duty.py::test_duty_amount_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_customs_duty.py::test_hs_chapter_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_customs_duty.py::test_parse_declaration_any_type_no_crash - ExceptionGroup: Hypothesis found 4 distinct failures. (4 sub-exceptions)
FAILED test_customs_duty.py::test_duty_amount_any_type_no_crash - ExceptionGroup: Hypothesis found 4 distinct failures. (4 sub-exceptions)
FAILED test_customs_duty.py::test_hs_chapter_any_type_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_customs_duty.py::test_parse_declaration_output_is_valid_or_none - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_customs_duty.py::test_duty_amount_output_is_valid_or_none - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_customs_duty.py::test_hs_chapter_output_is_valid_or_none - ValueError: invalid literal for int() with base 10: ''
FAILED test_customs_duty.py::test_known_bad_inputs_are_rejected - AttributeError: 'NoneType' object has no attribute 'split'
FAILED test_customs_duty.py::test_known_good_inputs - AssertionError: assert 0 == 1
========================================================== 11 failed, 3 passed in 6.15s ============================================================
```

Step 5 – Run the fixed script: `python customs_duty.py`

```
PS C:\Users\mukil\L6_L7_35_customs_duty\lab7> # Step 5: run the fixed script
PS C:\Users\mukil\L6_L7_35_customs_duty\lab7> python customs_duty.py
==== Running parse_declaration ====
Enter category:value (e.g. electronics:25000): electronics:25000
Category and Value: ('electronics', 25000.0)

==== Running duty_amount ====
Enter value@rate (e.g. 25000@20): 25000@20
Duty Amount: 5000.0

==== Running hs_chapter ====
Enter an HS code (e.g. 8517.12): 8517.12
Chapter: 85
PS C:\Users\mukil\L6_L7_35_customs_duty\lab7> # Type these three lines, pressing Enter after each:
PS C:\Users\mukil\L6_L7_35_customs_duty\lab7> #   electronics:25000
PS C:\Users\mukil\L6_L7_35_customs_duty\lab7> #   25000@20
PS C:\Users\mukil\L6_L7_35_customs_duty\lab7> #   8517.12
PS C:\Users\mukil\L6_L7_35_customs_duty\lab7> # → SCREENSHOT 3
PS C:\Users\mukil\L6_L7_35_customs_duty\lab7>
PS C:\Users\mukil\L6_L7_35_customs_duty\lab7> # Optional: run it again with bad input to show the fix
```

Invalid input is now handled gracefully (electronics25000 → Invalid input):

```
PS C:\Users\mukil\L6_L7_35_customs_duty\lab7> python customs_duty.py
==== Running parse_declaration ====
Enter category:value (e.g. electronics:25000): electronics25000
Category and Value: Invalid input

==== Running duty_amount ====
Enter value@rate (e.g. 25000@20):  25000@20
Duty Amount: 5000.0

==== Running hs_chapter ====
Enter an HS code (e.g. 8517.12):
Chapter: Invalid input
```

GPU TEMP	36 °C
GPU VOLTAGE	810 mV
GPU MEM CLK	1000 MHz
CPU UTIL	6 %

Step 6 – Re-run the tests against the fixed code: 14 passed

```
PS C:\Users\mukil\L6_L7_35_customs_duty\lab7> # Step 6: re-run the tests against the fixed code
PS C:\Users\mukil\L6_L7_35_customs_duty\lab7> python -m pytest test_customs_duty.py
============================================================ test session starts ============================================================
platform win32 -- Python 3.13.0, pytest-8.3.3, pluggy-1.5.0
rootdir: C:\Users\mukil\L6_L7_35_customs_duty\lab7
plugins: anyio-4.10.0, hypothesis-6.170.15, mock-3.14.0
collected 14 items

test_customs_duty.py ..............                                                                                                  [100%]

============================================================ 14 passed in 1.99s =============================================================
```

Bugs identified by fuzzing and how they were fixed

Function	Fuzz input	Original behaviour	Fix
all three	None, bytes	AttributeError / TypeError	isinstance(str) guard; return None
parse_declaration	'' , 'a:1:2'	ValueError (unpack)	require exactly one ':'
parse_declaration	':' , 'a:'	ValueError float('')	validate number and non-empty category
parse_declaration	'a:nan', 'a:inf', 'a:-5'	accepted silently	finite, non-negative ASCII numbers only
duty_amount	'', '@', '5@5@5'	ValueError	require exactly one '@'; validate both numbers
duty_amount	'5@nan', '-5@10', '1e308@1e308'	nan / negative / inf duty	reject; result must be finite
hs_chapter	'', '.', 'ab.12'	ValueError int('')	need 2 ASCII digits
hs_chapter	'5', '00.1', '98.1', '-1.2', non-ASCII digits	wrong chapter or accepted	chapter limited to 1–97, ASCII digits only
hs_chapter	' 0101.21 '	returned 0 (whitespace)	strip input first

Commits

Lab 7: add Hypothesis fuzz tests for all three functions (5258c58) · Lab 7: harden customs_duty.py against invalid input found by fuzzing (26cdd85)